REASON FOR VISIT: Primary couple infertility for 18 months and dull left scrotal dragging discomfort aggravated by prolonged standing.

PATIENT EVALUATION: A 31-year-old male and his 29-year-old spouse present for male factor infertility evaluation. The female evaluation by her reproductive endocrinologist confirmed regular ovulatory cycles and patent fallopian tubes on hysterosalpingography. The male patient notes an aching ache in his left hemiscrotum at the end of the day.

PHYSICAL EXAMINATION:

Genitalia: Phallus normal. Bilateral testes descended into the scrotum. Right testicular volume is 18 mL; left testicular volume is 13 mL (mild left testicular hypotrophy).

Palpation in upright position without Valsalva: Visual and palpable prominence of dilated, tortuous pampiniform venous plexus on the left resembling a 'bag of worms' (Grade III left varicocele). Right side is normal without varicocele.

DIAGNOSTIC TESTING:

1. Semen Analysis (two consecutive samples following 3 days of abstinence):

- Volume: 2.8 mL (normal > 1.5)

- Concentration: 9.4 million/mL (low, normal > 15 million/mL)

- Total Motility: 24% progressive motility (low, normal > 40%)

- Strict Kruger Morphology: 2% normal forms (low, normal > 4%)

- Interpretation: Oligoasthenoteratozoospermia (OATS).

2. Scrotal Color Doppler Ultrasound: Confirms left varicocele with multiple dilated veins measuring 3.8 mm at rest, increasing to 4.9 mm with retrograde venous flow lasting >2 seconds during Valsalva maneuver.

3. Endocrine Panel: Total testosterone 420 ng/dL, FSH mildly elevated at 8.2 mIU/mL, LH 4.1 mIU/mL.

RECOMMENDATIONS:

1. Clinically significant Grade III left varicocele associated with left testicular volume loss and abnormal semen parameters.

2. Discussed treatment options: Proceed with left subinguinal microscopic varicocelectomy.

3. Surgical rationale: High-ligation of internal spermatic veins with preservation of internal spermatic arteries and lymphatic channels improves semen quality in 60-70% of men.